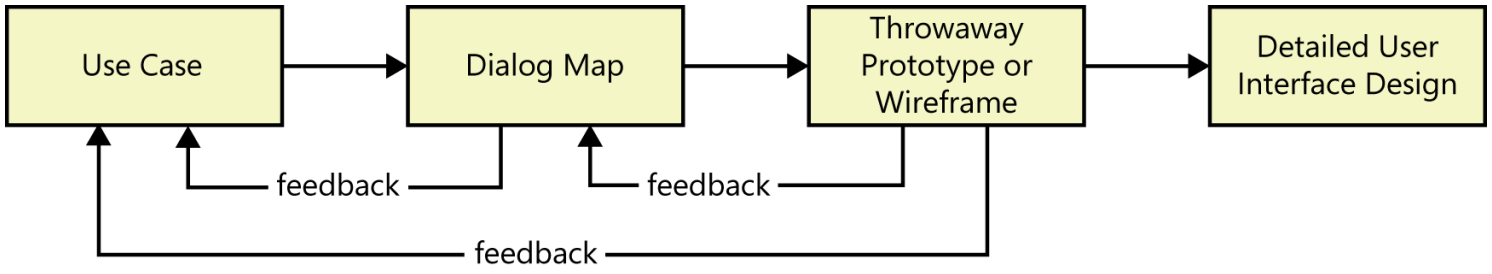
L4 - UX & Prototyping

Topics Covered

- ✧ Prototyping
- ✧ UI/UX design

Working with prototyping

- Always need **executable prototype** to resolve requirements uncertainties?
  - A **paper prototype** (sometimes called a low-fidelity prototype) is a cheap, fast, and low-tech way to explore how a portion of an implemented system might look (Rettig 1994).
  - A similar deliverable is called a **storyboard**.
- One possible sequence of development activities that moves from use cases to detailed user interface design:



(Ref: Chapter 15 - Wiegers, Karl E., and Joy Beatty. *Software requirements*. Pearson Education, 2013.)

Use cases

The Example for PearlsFromSand.com

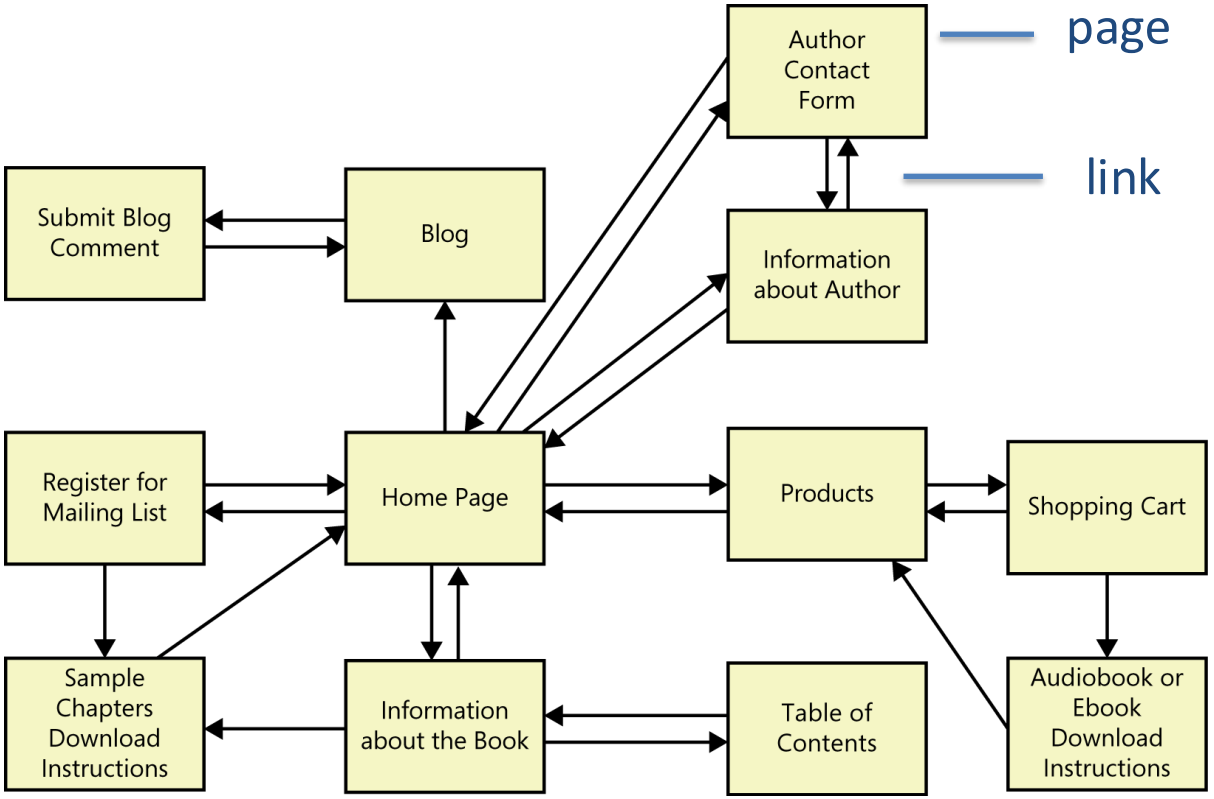
User class	Use case
Visitor	Get Information about the Book Get Information about the Author Read Sample Chapters Read the Blog Contact the Author
Customer	Order a Product Download an Electronic Product Request Assistance with a Problem
Administrator	Manage the Product List Issue a Refund to a Customer Manage the Email List

Some use cases for PearlsFromSand.com

Dialog Map

The Example for PearlsFromSand.com

- A dialog map that illustrates a **conceptual page architecture**.
- Discover new actions
- Find ways to simplify and streamline the user's experience

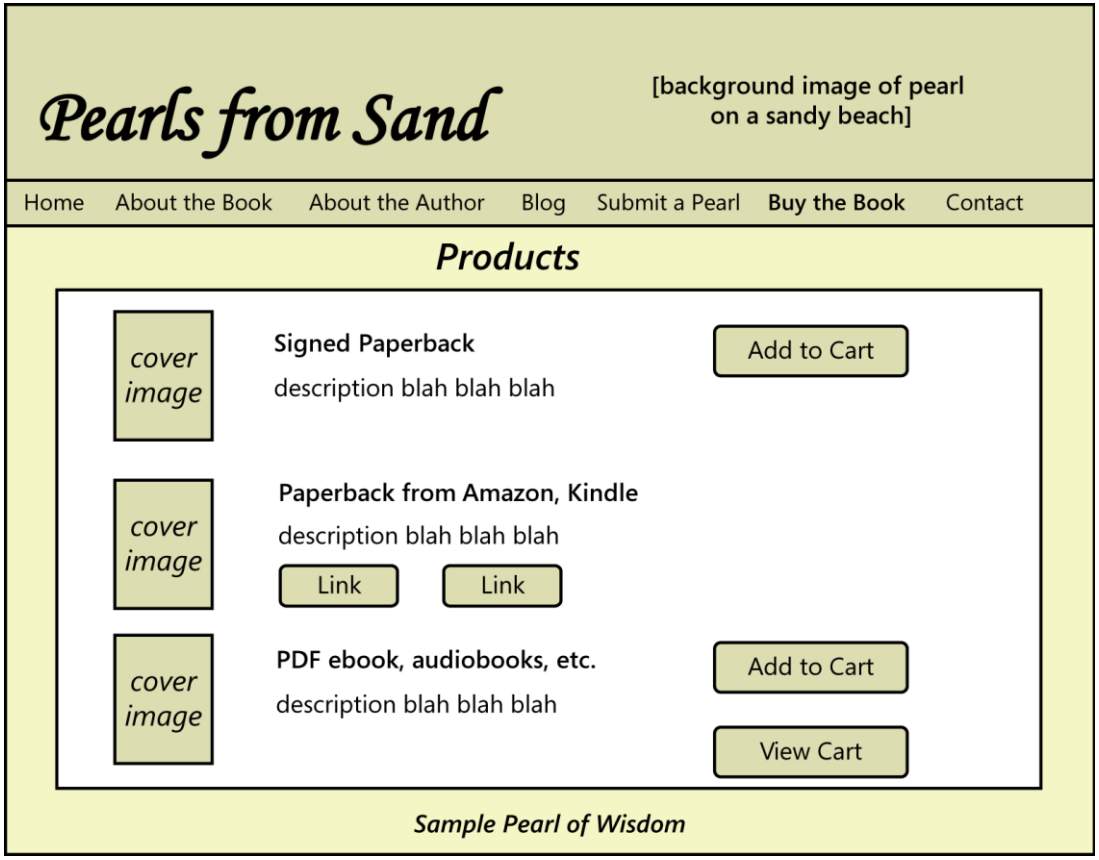


Partial dialog map for PearlsFromSand.com

Paper prototypes / Wireframe

The Example for PearlsFromSand.com

- Can be a **hand-drawn sketch on paper**, a simple line drawing, or a mock-up created with a dedicated prototyping or visual design tool
- Understand what page layout and would make the pages easy to understand and use



Sample wireframe of one page for PearlsFromSand.com by using Powerpoint in just a few mins

Detailed User Interface Design

The Example for PearlsFromSand.com



A final implemented page from PearlsFromSand.com

Prototyping: Success Factors and Risks

- **Include prototyping tasks** in your project plan.
- State the **purpose** of each prototype before you build it, and **explain** what will happen with the outcome.
- Plan to develop **multiple prototype**.
- Create throwaway prototypes as **quickly and cheaply** as possible. **Don't include** input data validations, defensive coding techniques, error-handling code, or extensive code doc.
- **Don't** prototype requirements that you **already understand**, except to explore design alternatives.
- **Use plausible data** in prototype screen displays and reports.
- **Don't** expect a prototype to **replace written requirements**.

Risks

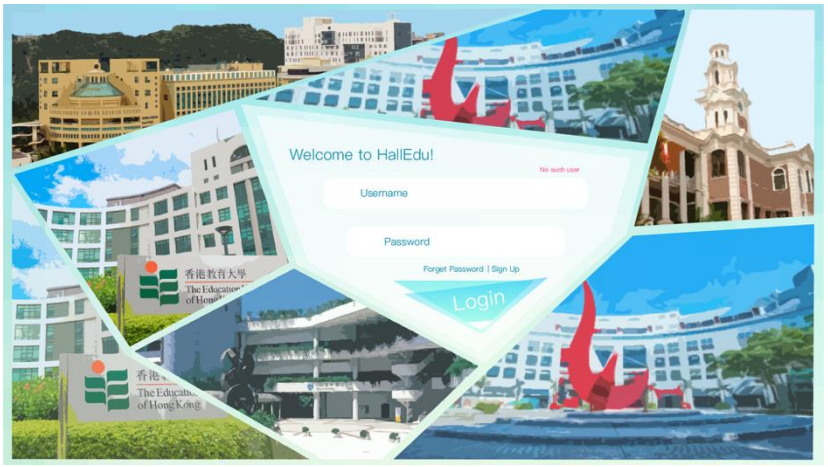
Pressure to release — "it looks almost done!" A throwaway prototype is never production-ready.

Distraction by details — users fixate on how screens look, not on what the system must do.

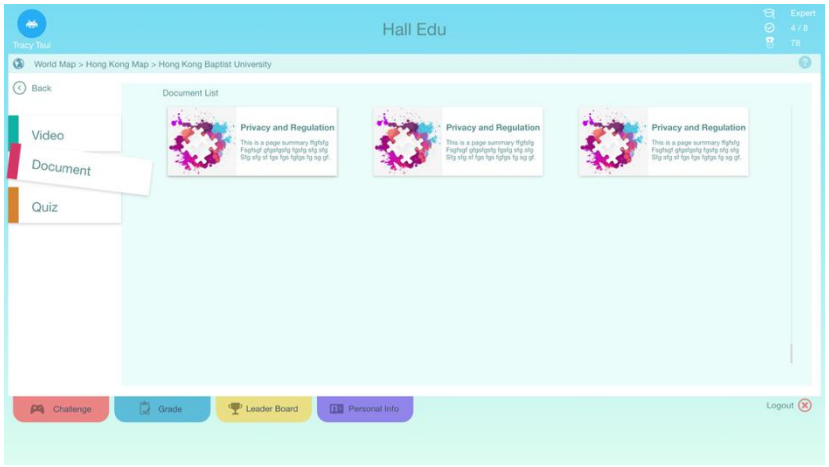
Unrealistic performance — the mock-up feels faster than the real product will be.

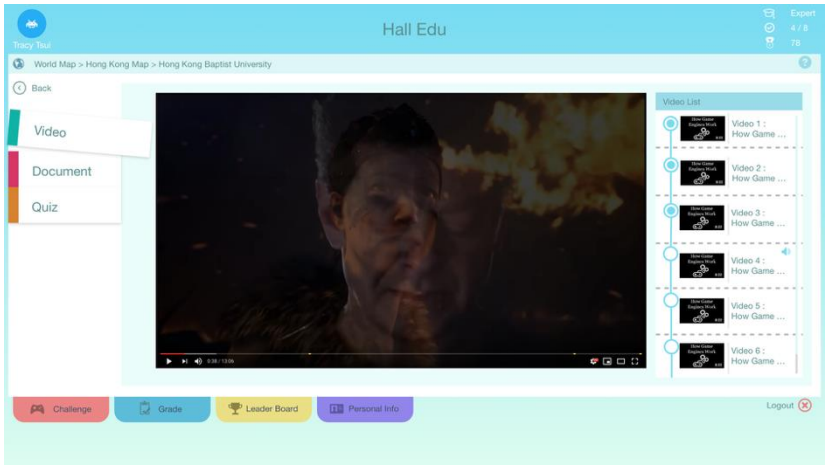
Excessive effort — a prototype is an experiment: keep it quick and cheap.

User Interface Design of Hall Mission: The Gamified Q&A System, by Adobe XD



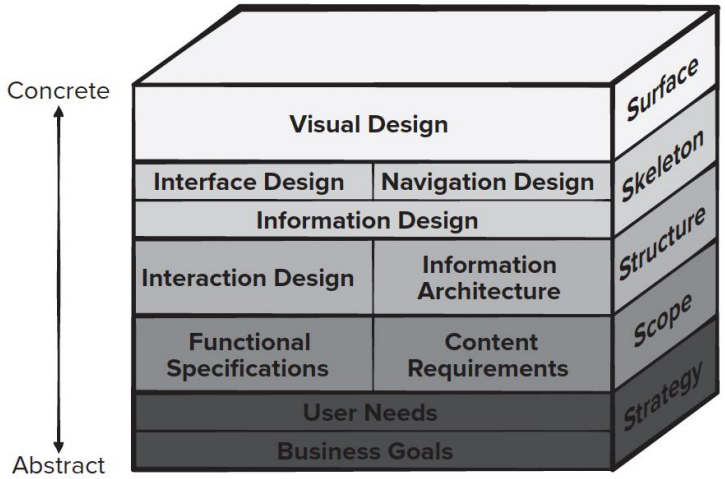






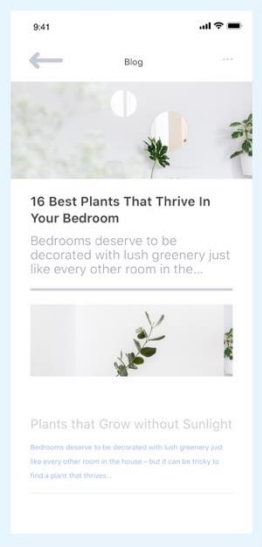
User Experience Design

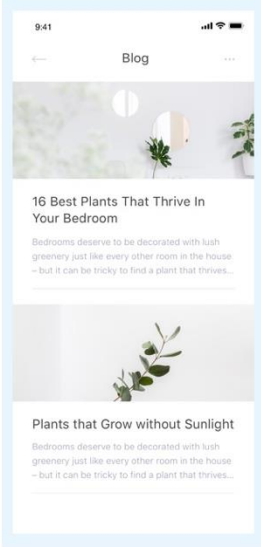
- User Experience (UX) design is **a set of incremental process activities** that help the development team and the project stakeholders focus on providing a **positive experience for users** of the software product.
- UX design is **broader** than User Interface (UI) design and usability or accessibility engineering.
- To make the task of crafting a positive user experience more **manageable**, Garret [Gar10] suggests breaking it down into component elements: strategy, scope, structure, skeleton, and surface.

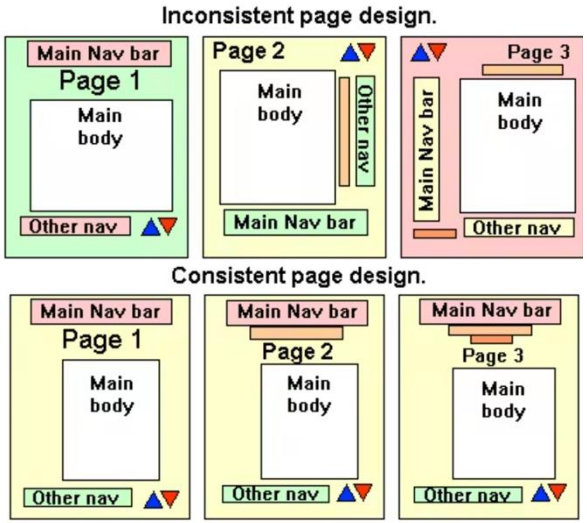


User Experience Design Elements

Typical Design Errors (1)

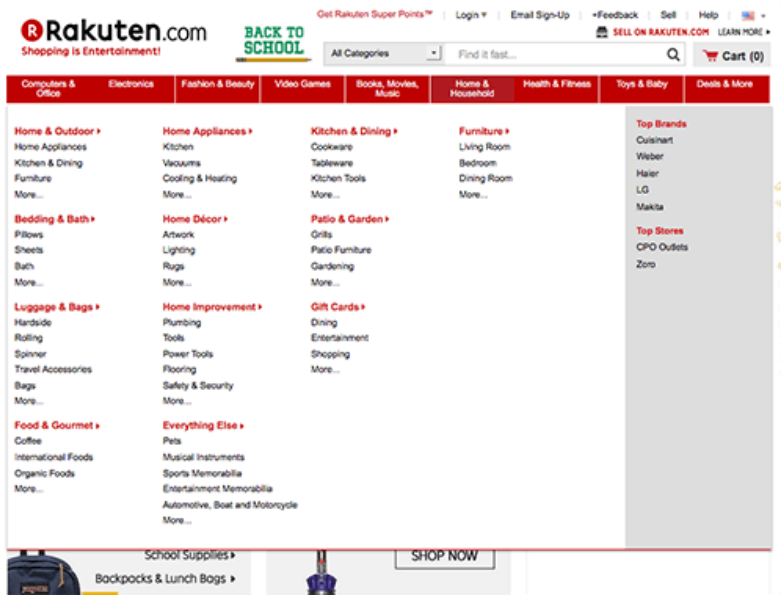






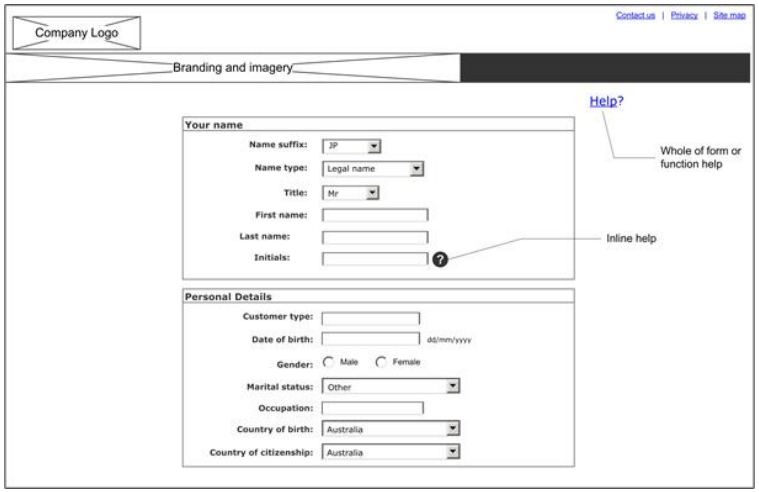
Typical Design Errors (2)





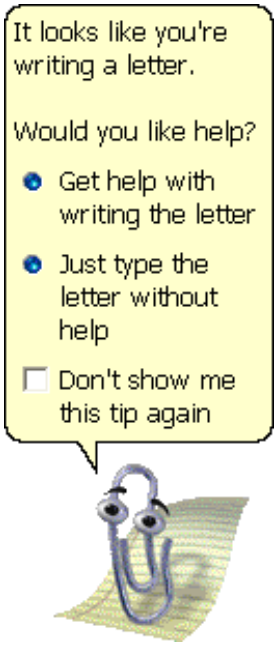
Typical Design Errors (3)





Typical Design Errors (4)

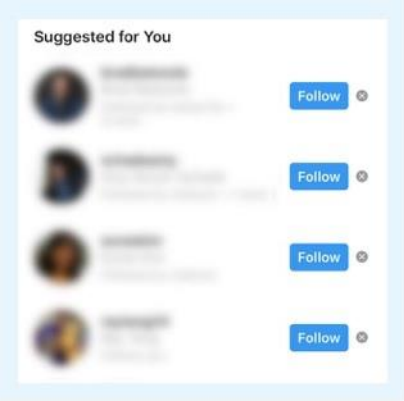


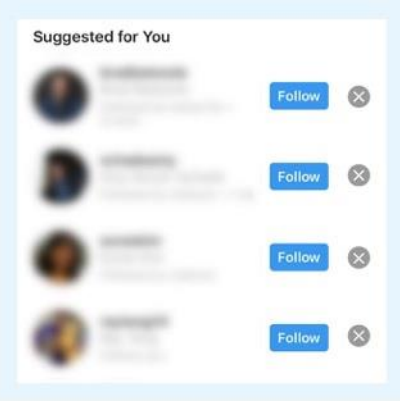


Typical Design Errors (5)

- ✧ Poor response
  - Slow page loading speed
  - No response to selected web page click buttons or fields

Typical Design Errors (6)

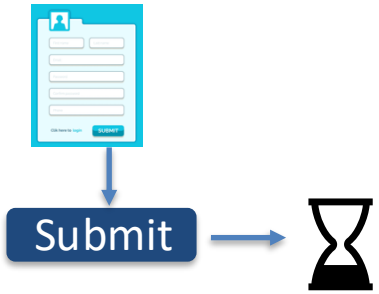


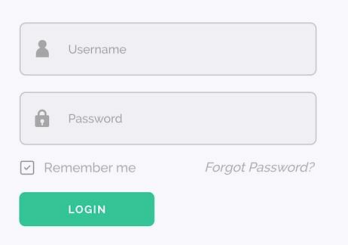




Usability Guidelines (1)

- **Anticipation** - An application should be designed so that it anticipates the **user's next move**.
- **Communication** - The interface should communicate the **status** of any activity initiated by the user.
- **Consistency** - The use of navigation controls, menus, icons, and aesthetics (e.g., color, shape, layout) should be consistent throughout.
- **Controlled autonomy** - The interface should facilitate user movement throughout the application, but it should do so in a manner that **enforces navigation conventions** that have been established for the application.
- **Efficiency** - optimize the user's work efficiency, **NOT** the efficiency of the developer who designs/builds it or the client-server environment that executes it.





Usability Guidelines (2)

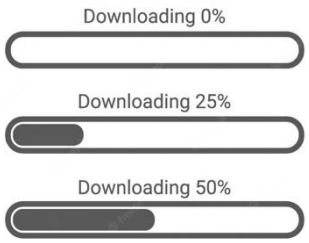
- **Flexibility** - Flexible enough to enable some users to accomplish tasks directly and others to explore the application in a somewhat random fashion.



- **Focus** - The interface should stay focused on the user task(s) at hand.
- **Human interface objects** - A vast library of reusable human interface objects has been developed for Web and mobile apps. Use them.



- **Latency reduction** - Rather than making the user wait for some internal operation to complete (e.g., downloading a video), the app should use **multitasking in a way** that lets the user proceed with work as if the operation has been completed.



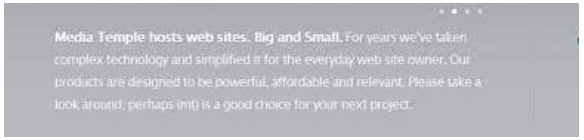
- **Learnability** - Minimize learning time, and once learned, to minimize relearning required when the app is revisited.

Usability Guidelines (3)

- **Metaphors** - An interface that uses an **interaction metaphor** is easier to learn and easier to use, as long as the metaphor is appropriate for the application and the user.
- **Readability** - All information presented through the interface should be **readable** by young and old.
- **Track state** - When appropriate, the state of the user interaction should be tracked and **stored** so that a user can logoff and return later to pick up where he/she left off.
- **Visible navigation** - A well-designed interface provides “the illusion that users are in the same place, with the work brought to them.”



Shopping cart





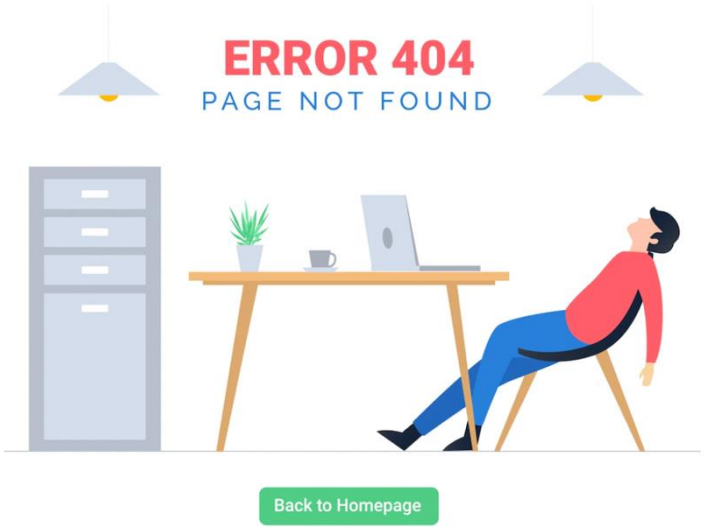


Accessibility Guidelines

- ✧ Application Accessibility
- ✧ Response Time
- ✧ Help Facilities
- ✧ Error Handling
- ✧ Menu and Command Labeling
- ✧ Internationalization



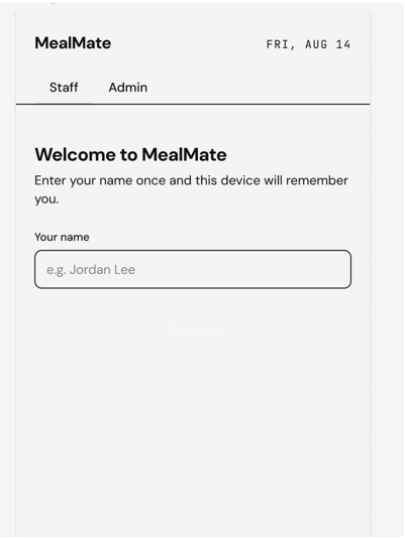






AI in UI Design: Draft, then Refine

- AI tools generate first-draft screens from a text prompt - in W4 you will use Figma's own, Figma Make.
- The draft is a starting point - apply THIS lecture to it:
  - Check against the typical design errors you just learned.
  - Check usability guidelines: consistency, feedback, error prevention.
  - Check accessibility: contrast, touch targets, text size.
- W4: hand-build one screen, let Make draft the app, move it all into Figma Design, and send a prototype link.



W4: Figma Make shipped a Continue button nobody can read — the Readability / Contrast check in this lecture catches it (own screenshot).

References

- Pressman - Software Engineering: A Practitioner's Approach, 7/e (McGraw-Hill)
- Nielsen Norman Group - 10 Usability Heuristics
- W3C - WCAG accessibility guidelines
- Figma Learn |Figma for Education (free for students)